LESSON 6.14

The Distributive Property

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use the distributive property to expand and factor expressions.

WHY THIS MATTERS You're selling concert bundles at the music studio. Each bundle includes a ticket ($15) and a snack ($5). If 3 people each buy a bundle, you can calculate the total as 3(15 + 5) OR as 3 × 15 + 3 × 5. Either way, the total is the same — $60! This is the distributive property.

REMEMBER

The Distributive Property

The Distributive Property. Formula: a(b + c) = ab + ac | a(b − c) = ab − ac The distributive property means you can multiply a number by everything inside the parentheses, one piece at a time: a(b + c) = ab + ac.

1. Expand: Multiply the outside factor by each term inside parentheses.
2. Factor: Find the GCF of terms and pull it outside: ab + ac = a(b + c).
3. Visual: Represent as the area of partitioned rectangles

Words: **The Distributive Property *(La propiedad distributiva)*** **Distributive Property *(Propiedad distributiva)*** **Factor *(Factor)*** **Expand *(Desarrollar)*** **Equivalent *(Equivalente)***

1 **Circle the letter of the best answer. Show how you know.**

What is the prime factorization of 24?

(A) 2 × 2 × 2 × 3
(B) 2 × 12
(C) 3 × 8
(D) 4 × 6

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which number is prime?

(A) 19
(B) 21
(C) 27
(D) 33

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A factor tree for 36 ends with which prime factors?

(A) 2 × 2 × 3 × 3
(B) 2 × 18
(C) 3 × 12
(D) 6 × 6

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

In the prime factorization of 50, how many times does 5 appear?

(A) Once
(B) Three times
(C) Twice
(D) Not at all

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A music studio is ordering supplies for 8 practice rooms. Each room needs a music stand ($12) and a stool ($9). The total cost can be written as 8(12 + 9). Distribute: 8(12 + 9) = ?

(A) 8 × 12 + 8 × 9
(B) 8 × 12 + 9
(C) 8 + 12 × 9
(D) 8 × 21 × 8

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A music studio is ordering supplies for 8 practice rooms. Each room needs a music stand ($12) and a stool ($9). The total cost can be written as 8(12 + 9). What is the total cost for 8 practice rooms?

(A) $21
(B) $96
(C) $105
(D) $168

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you see when you expand an expression using the distributive property?

SENTENCE STARTER When I distribute, I multiply the outside factor ___ by each term ___. The operation inside the parentheses ___ stays the same.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each a(b + c) light cue to its ab + ac expanded form.

_____ 2(b + c)	(A) 2b + 2c
_____ 6(x + 2)	(B) 6x + 12
_____ 3(m + 5)	(C) 3m + 15
_____ 5(p + 1)	(D) 5p + 5

2 **Write the name of the correct group on each line.**

Sort each light-cue expansion: was the factor distributed correctly to BOTH terms, or only to one?

Groups: Distributed correctly | Distributed incorrectly

_____ 2(x + 6) = 2x + 12

_____ 4(n + 3) = 4n + 3

_____ 5(y + 2) = 5y + 10

_____ 3(a + 7) = 3a + 7

_____ 6(m + 1) = 6m + 6

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A music studio is ordering supplies for 8 practice rooms. Each room needs a music stand ($12) and a stool ($9). The total cost can be written as 8(12 + 9). A student wrote 8(12 + 9) = 8 × 12 + 9 = 105. What went wrong?

(A) They added instead of multiplying

(B) They used the wrong prices

(C) They forgot to multiply the 9 by 8 as well

(D) They distributed too many times

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Light Cue 1: Expand 3(x + 4) for the stage-light controller.

(A) 3x + 12

(B) 3x + 4

(C) 3x + 7

(D) x + 12

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Light Cue 2: Expand 5(2 + n) to set two spotlight banks.

(A) 7 + 5n

(B) 10 + 5n

(C) 10 + n

(D) 10 + 5

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Light Cue 3: Expand 4(a + 2) for the back-row beam.

(A) 4a + 8

(B) 4a + 2

(C) 4a + 6

(D) a + 8

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you see when you expand an expression using the distributive property?

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